

EPPS 6302 Methods of Data Collection and Production:

Assignment 3

ACS via tidycensus

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What you will produce

A tidy ACS dataset pulled with `tidycensus`, one choropleth map, one table, and a short methods note that takes the margins of error seriously.

Objective

Use `tidycensus` to retrieve ACS 2024 5-year estimates, build a tidy dataset, and produce **one map** and **one table** with a short interpretation.

1. Setup

- Obtain a Census API key — https://api.census.gov/data/key_signup.html
- Install the key **once, from the console**, then never mention it again:

```
census_api_key("YOUR_KEY", install = TRUE) # console only — not in your .qmd
```

- Install and load: `tidycensus`, `tigris`, `sf`, `dplyr`, `tidyr`, `ggplot2`, `readr`.

The key never goes in the document. `install = TRUE` writes it to your `.Renv` once; every later session picks it up by itself. `install = FALSE` means the key sits in the file — and this file is going onto your public website. Add `.Renv` to `.gitignore`.

2. Choose a geography and variables

- Geography:** one of `state`, `county`, `tract`, or `block group` (within a state).
- Variables:** at least two. For example `B19013_001` (median household income) and `B17001_002` (people below poverty).
- If you use a `count` such as `B17001_002`, also pull its denominator `B17001_001` so you can report a **rate**. A count map is mostly a population map.
- Search the codes with `load_variables(2024, "acs5")`.

3. Download, with geometry

- Use `get_acs()` with your geography and variables, `geometry = TRUE`.
- Keep the **estimate and the margin of error**. Every ACS number is an estimate from a sample; the MOE is not decoration.

Starter code — edit to your choices

The same code ships as `assignment03_starter.qmd`. Open that file and work in it.

```
# Packages
library(tidycensus)
library(tigris)
library(sf)
library(dplyr)
library(ggplot2)
library(readr)

options(tigris_use_cache = TRUE)

# 1) API key — run ONCE in the console, never in this file
# census_api_key("YOUR_CENSUS_KEY", install = TRUE)

# 2) Explore variables (no `cache` argument — deprecated and ignored)
vars <- load_variables(2024, "acs5")
vars |> dplyr::filter(grepl("^B19", name)) |> dplyr::slice_head(n = 10)

# 3) Parameters (EDIT ME)
state_abbr <- "TX"
geo_level <- "county" # options: state, county, tract, block group
my_vars <- c(income = "B19013_001", # median household income
             pov = "B17001_002", # people below poverty
             povden = "B17001_001") # poverty universe = denominator
year_acs <- 2024
survey <- "acs5"

# 4) Download, wide — one row per area, and the sf class survives
acs_wide <- get_acs(
  geography = geo_level,
  variables = my_vars,
  state = state_abbr,
  year = year_acs,
  survey = survey,
  geometry = TRUE,
  output = "wide"
)

# 5) Derive a rate, and carry its margin of error through
acs_wide <- acs_wide |>
  mutate(pov_rate = povE / povdenE,
         pov_moe = moe_prop(povE, povdenE, povM, povdenM))

# 6) Map (edit titles/theme)
ggplot(acs_wide) +
  geom_sf(aes(fill = incomeE), color = NA) +
  scale_fill_viridis_c(name = "Median HH Income") +
  labs(title = paste0("ACS ", year_acs, " 5-year: Median Income — ",
                    state_abbr, " (", geo_level, ")"),
       caption = "Source: U.S. Census Bureau via tidycensus") +
  theme_minimal() +
```

```

theme(text = element_text(family = "Palatino"), # drop if not installed
      plot.title = element_text(size = 14, face = "bold"),
      plot.caption = element_text(size = 8),
      legend.position = "inside",
      legend.position.inside = c(0.2, 0.15))

# 7) Table (top/bottom by poverty rate, with MOE)
top10 <- acs_wide |>
  st_drop_geometry() |>
  arrange(desc(pov_rate)) |>
  select(NAME, pov_rate, pov_moe, povE, povM) |>
  slice_head(n = 10)

bottom10 <- acs_wide |>
  st_drop_geometry() |>
  arrange(pov_rate) |>
  select(NAME, pov_rate, pov_moe, povE, povM) |>
  slice_head(n = 10)

top10
bottom10

# 8) Save outputs
write_csv(st_drop_geometry(acsc_wide),
          paste0("acs_", state_abbrev, "_", year_acs, ".csv"))

```

4. Deliverables

- Map** — a choropleth of one variable, with a clear legend and title.
- Table** — the top and bottom ten areas by the other variable, with the margin of error beside each estimate.
- Uncertainty check** — name one pair of areas in your table whose confidence intervals overlap. What does that do to the ranking you just published?
- Methods note** (≤ 200 words) — your choices of geography and variables, and what the MOEs do to your interpretation.
- Publish as `assign03.qmd` under **Assignments** on your website

Notes

- Prefer **rates** to counts where a denominator exists, and propagate the MOE with `moe_prop()`, `moe_ratio()` or `moe_sum()` rather than dropping it.
- Mapping tracts or block groups for a whole state is slow and unreadable — filter to one county or metro area first.

Two different caches. `options(tigris_use_cache = TRUE)` caches map geometry and is still correct. The `cache` argument of `load_variables()` is not: current tidycensus reports it “*deprecated and ignored — variable metadata is now loaded from the Census API without writing to a local cache.*” Drop it.